

Ques: Print the given pattern

```

1
0 1
1 0 1
0 1 0 1

```

$n=4$

```

      1 2 3 4 5 → 1
1      |

```

```

2 0 1

```

```

3 1 0 1

```

```

4 0 1 0 1

```

```

5 1 0 1 0 1

```

↓ $n=5$
i

if $(i+j)$ is even $\rightarrow 1$

else $\rightarrow 0$

Binary Triangle

Summary Notes

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★ Method-1:

Ques: Print the given pattern

1 2 3 4 → ↓

```

1 # # # *
2 # # * *
3 # * * *
4 * * * *

```

└ n=4

↓

if $(i+j) > n$ cout << "*",
else cout << "#",

Star Triangle Vertically Flipped

Method-2



Earn₹s

Ques: Print the given pattern

```
1  _ _ _ *
2  _ _ * *
3  _ * * *
4 * * * *
```

=

```
1  _ _ _
2  _ _
3  _
4
```

Spaces

+

```
1  *
2  * *
3  * * *
4  * * * *
```

Stars

Star Triangle Vertically Flipped

Ques: Print the given pattern

```

1   _ _ _ 1
2   _ _ 1 2
3   _ 1 2 3
4 1 2 3 4
  
```

$n=4$

=

```

1   _ _ _
2   _ _
3   _
4
  
```

+

```

1 1
2 1 2
3 1 2 3
4 1 2 3 4
  
```

Code → HW

Number Triangle Vertically Flipped

HW: Print the given pattern

```
1  _ _ _ A
2  _ _ B B
3  _ C C C
4  D D D D
```

Method-1 & 2 both

Alphabet Triangle Vertically Flipped

Ques: Print the given pattern

```

1  _ _ _ * * * *
2  _ _ * * * *
3  _ * * * *
4 * * * *
  
```

$n=4$

=

```

1  _ _ _
2  _ _
3  _
4
  
```

+

```

1  x x x x
2  x x x x
3  x x x x
4  x x x x
  
```

Rhombus

Q Print the pattern

1 2 3 4 5 $\rightarrow j$

1 *

2 * * *

3 * * * * *

$\downarrow$
1

$n=3$

$$j \leq 2i - 1$$

1 2 3 4 5 6 7 8 9 $\rightarrow j$

1 *

2 * * *

3 * * * * *

4 * * * * * * *

5 * * * * * * * *

$\downarrow$
1

$n=5$



Earn ₹s

HW Ueta Pyramid

$n=4$

1	*	*	*	*	*	*	*
2	—	*	*	*	*	*	
3	—	—	*	*	*		
4	—	—	—	*			

$$nst = 2^n - 1$$

$$nsp = 0$$

$$nst -= 2$$

$$nsp++$$

✖ ✖
HW

Hollow Diamond

$n=4$

```
  _ _ _ *
 _ _ * _ *
_ * _ _ _ *
* _ _ _ _ _ *
_ * _ _ _ *
_ _ * _ *
_ _ _ _ *
```

* Ques: Print the given pattern

```

1  _ _ _ *
2  _ _ * * *
3  _ * * * * *
4  * * * * * * *
  
```

n=4

```

=
1  _ _ _
2  _ _
3  _
4  .

+
1  *
2  * * *
3  * * * * *
4  * * * * * * *
  
```

Pyramid

Method-2



Earn₹s

Ques: Print the given pattern

```
1  _ _ _ *
```

```
2  _ _ * * *
```

```
3  _ * * * * *
```

```
4 * * * * * *
```

$n=4$

$nsp = n - 1$

$nsp -= 1$

$nst = 1$

$nst += 2$

Pyramid

Ques: Print the given pattern

$n = 1$ $n = 2$ $n = 3$

```

1  _ _ _ *
2  _ _ * * *
3  _ * * * * *
4  * * * * * * *

```

```

1  _ * * * * *
2  _ _ * * *
3  _ _ _ *

```

$n = 4$

$nsp = n - 1$
 $nst = 1$

$nsp = 1$
 $nst = 2n - 3$

Diamond

Method-2

Ques: Print the given pattern

1	—	—	—	*						
2	—	—	*	*	*					
3	—	*	*	*	*	*				
4	*	*	*	*	*	*	*			
5	—	*	*	*	*	*				
6	—	—	*	*	*					
7	—	—	—	*						

nst += 2
nsp --

nst -= 2
nsp ++

n=4

Diamond

Ques: Print the given pattern

```

* * * * *
* * * *   * * *
* * *     * * *
* *       * *
*         * *
          *
  
```

$n = 5$

$n = 2$

```

* * *
*   *
  
```

$n = 3$

```

* * * * *
* *   * *
*     *
  
```

Bridge

Ques: Print the given pattern

↖ along se $(2n-1)$ stars ko print

```

* * * * *
1 * * * _ * * *
2 * * * _ _ _ * *
3 * * _ _ _ _ *
4 * _ _ _ _ _ _ *

```

(1 Loop me 3 loop)

$n=5$

```

1 x x x x
2 x x x
3 x x
4 x
+
1 _
2 _ _ _
3 _ _ _ _
4 _ _ _ _ _
+
1 x x x x
2 x x x
3 x x
4 x

```

Bridge

Ques: Print the given pattern

```
1 1 1 1 1 1 1
1 2 2 2 2 2 1
1 2 3 3 3 2 1
1 2 3 4 3 2 1
1 2 3 3 3 2 1
1 2 2 2 2 2 1
1 1 1 1 1 1 1
```

Number Spiral